

SOC THREAT DETECTION IN A SEGMENTED HEALTHCARE NETWORK

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EXECUTIVE SUMMARY

MedSecure Health Solutions, a healthcare technology provider managing patient records and telemedicine services, noticed unusual activity targeting its EHR server, showing lapses in monitoring and internal controls.

To strengthen security, the network was segmented into WAN, LAN, and DMZ zones, and tools such as firewalls, centralized logging, and traffic-analysis systems were deployed for improved visibility. Simulated attacks, including repeated login failures and internal scanning, were conducted to test detection and response capabilities.



PROJECT INTRODUCTION

1

MedSecure manages sensitive health and telemedicine data.

2

Recently, unusual activity targeting the EHR server in the DMZ raised concerns about weak visibility and gaps in internal controls. These events showed that the existing setup was not fully capable of detecting or understanding potential threats.

3

SOC Capstone aimed to test detection, analysis, and response. and focus on network segmentation and centralized monitoring.

METHODOLOGY

1

Designed segmented network: WAN, LAN, DMZ.

Configured firewall rules and traffic restrictions.

2

Installed Wazuh Manager and agents.

Simulated attacks (scanning, brute-force).

3

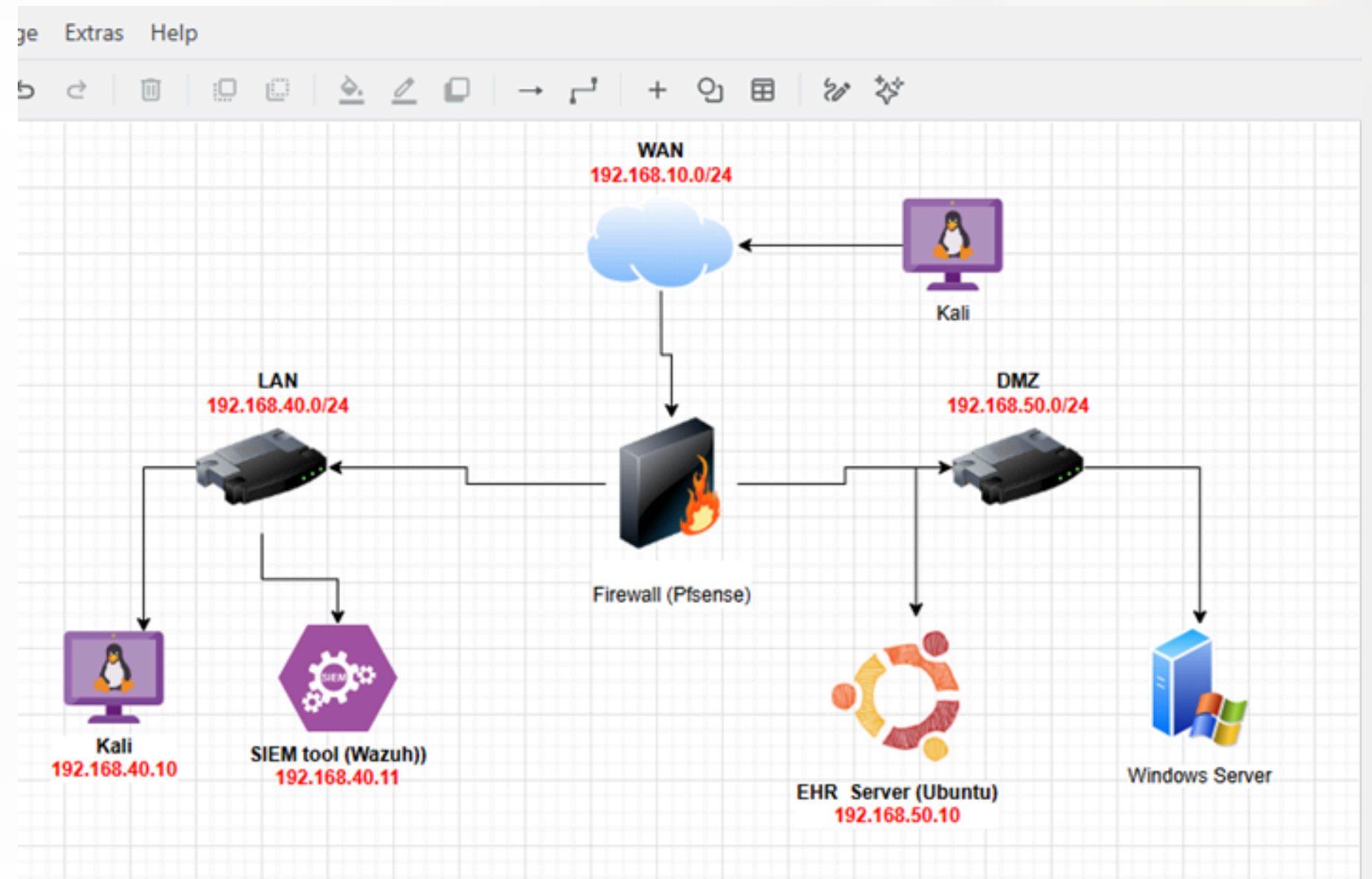
Captured traffic with Wireshark.

Correlated logs and validated alerts.

PHASE 1 NETWORK SEGMENTATION

A network diagram was created in Draw.io showing WAN, LAN, DMZ, key hosts, and the firewall to visualize traffic flows and boundaries.

In pfSense, the network was segmented to enforce clear boundaries between the zones. The WAN network, named MedSecure, was configured as the main source of internet connectivity, with DHCP enabled to automatically assign IP addresses. WAN, LAN, and DMZ interfaces were assigned to separate adapters (WAN: Adapter 1, LAN: Adapter 2, DMZ: Adapter 3), with DHCP enabled on the WAN.



PHASE 1

```
pfsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MedSecure ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.10.4/24
LAN (lan) -> em1 -> v4: 192.168.40.1/24
DMZ (opt1) -> em2 -> v4: 192.168.50.1/24

0) Logout / Disconnect SSH      9) pfTop
1) Assign Interfaces            10) Filter Logs
2) Set interface(s) IP address 11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system               14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option:
Message from syslogd@MedSecure at Dec 10 00:14:21 ...
php-fpm[435]: /index.php: Successful login for user 'admin' from: 192.168.40.11
(Local Database)

Message from syslogd@MedSecure at Dec 10 01:26:32 ...
php-fpm[434]: /interfaces_assign.php: Successful login for user 'admin' from: 192.168.40.11
(Local Database)
```

pfSense - Settings

Basic Expert

General System Display Storage Audio **Network** Serial Ports USB Shared Folders User Interface

Search settings

Network

Adapter 1 Adapter 2 Adapter 3 Adapter 4

☒ Enable Network Adapter

Attached to: NAT Network

Name: MedSecure-WAN

Adapter Type: Intel PRO/1000 MT Desktop (B2545FA)

Promiscuous Mode: Deny

MAC Address: C8002787CC96

☒ Virtual Cable Connected

Serial Ports

Port 1 Port 2 Port 3 Port 4

OK Cancel Help

19

✖

0/504 B

IPv4 *

LAN subnets

*

DMZ subnets

*

*

none

✓

2/320.66 MiB

IPv4 *

LAN subnets

*

*

*

*

none

Default allow LAN to any rule

✓

0/0 B

IPv6 *

LAN subnets

*

*

*

*

none

Default allow LAN IPv6 to any rule

↑

Add

↓

Add

🗑

Delete

🔕

Toggle

📄

Copy

💾

Save

+

Separator

PHASE 2

ATTACK SIMULATION & TRAFFIC CAPTURE

```
(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this
is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2023-12-09 10:54:36
[DATA] max 4 tasks per 1 server, overall 4 tasks, 132 login tries (l:11/p:12), ~33 tries per task
[DATA] attacking ssh://192.168.50.10:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://medsecureadmin@192.168.50.10:22
[INFO] Successful, password authentication is supported by ssh://192.168.50.10:22
[22][ssh] host: 192.168.50.10 login: medsecureadmin password: admin
[22][ssh] host: 192.168.50.10 login: medsecure password: medsecure
[STATUS] 88.00 tries/min, 88 tries in 00:01h, 44 to do in 00:01h, 4 active
[22][ssh] host: 192.168.50.10 login: ubuntu password: ubuntu
[STATUS] attack finished for 192.168.50.10 (waiting for children to complete tests)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2023-12-09 10:56:06
```

In this phase, Kali Linux was used to simulate common attacker behaviors, including credential-guessing attempts and network scanning.

Wireshark was run on selected network interfaces to capture traffic during these simulated attacks. This allowed monitoring of SSH authentication attempts, scanning activity, and other suspicious packets generated by the testing processes.

```
(kali@kali)-[~]
$ sudo systemctl start ssh
[sudo] password for kali:

(kali@kali)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenSSH Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Tue 2023-12-09 00:53:11 EST; 0s ago
     Invocation: d0267e43a7f1489001fa14204cff4201
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 32279 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 32290 (sshd)
      Tasks: 1 (limit: 2110)
     Memory: 5.1M (peak: 5.6M)
        CPU: 52ms
    CGroup: /system.slice/ssh.service
            └─32290 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 09 00:53:10 kali systemd[1]: Starting ssh.service - OpenSSH Secure Shell server...
Dec 09 00:53:11 kali sshd[32290]: Server listening on 0.0.0.0 port 22.
Dec 09 00:53:11 kali sshd[32290]: Server listening on :: port 22.
Dec 09 00:53:11 kali systemd[1]: Started ssh.service - OpenSSH Secure Shell server.

(kali@kali)-[~]
$ nmap -help
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] [target specification]
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanner.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3], ... >: Exclude hosts/networks
```

PHASE 2

[illegible]

```
(kali@kali)-[~]
$ nmap 192.168.50.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:28 EST
Nmap scan report for 192.168.50.1
Host is up (0.00055s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap scan report for 192.168.50.10
Host is up (0.0016s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (2 hosts up) scanned in 9.65 seconds

(kali@kali)-[~]
$ nmap 192.168.50.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:31 EST
Nmap scan report for 192.168.50.10
Host is up (0.0021s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds

(kali@kali)-[~]
$ hydra --help
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organs
is non-binding, these *** ignore laws and ethics anyway).
```

PHASE 3

SECURITY MONITORING AND RESPONSE

Wazuh detected failed logins and scanning.

High-level alerts triggered during brute-force attempt.

Automatic firewall blocking stopped the attacker.

Demonstrated strong detection and automated defense.



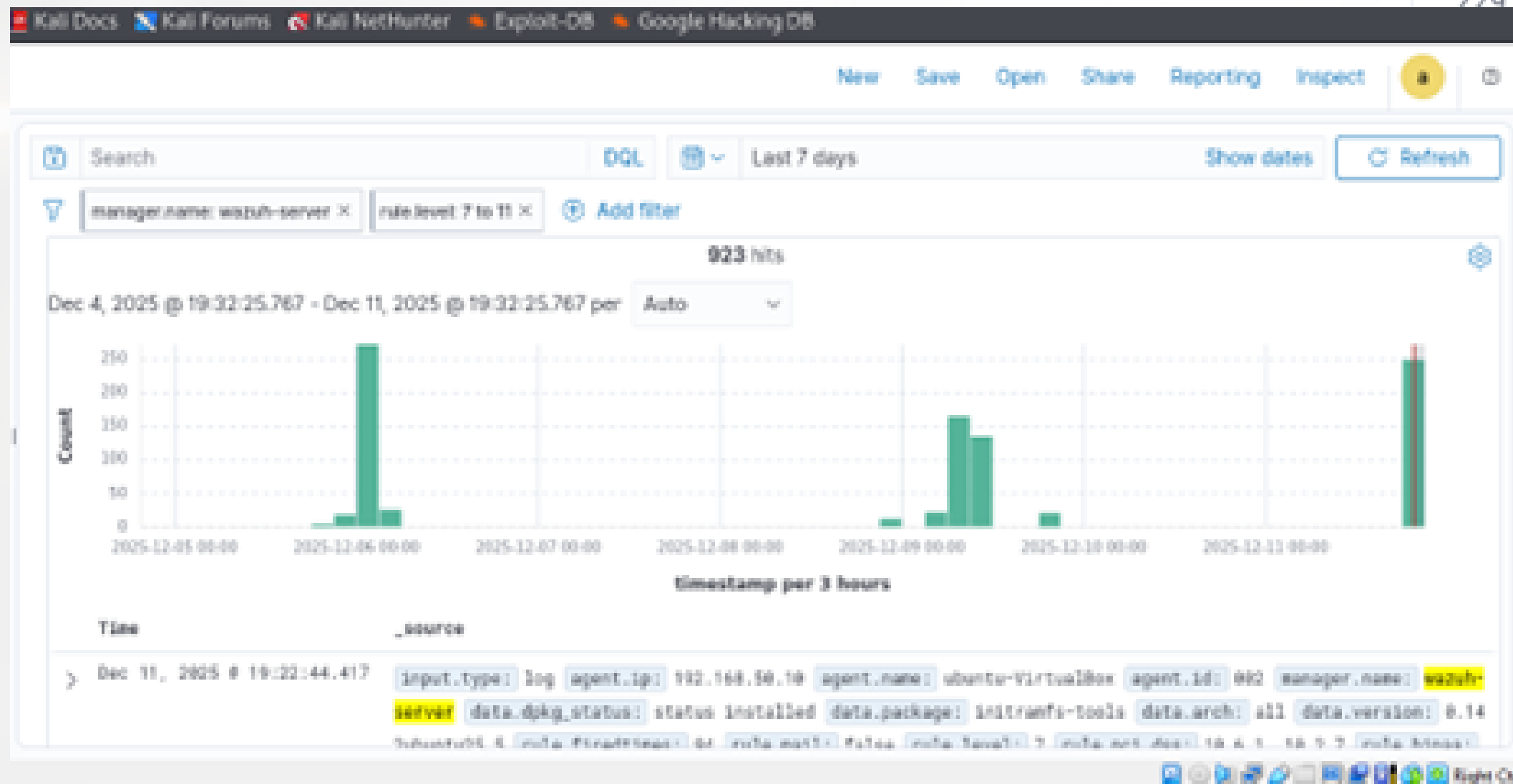
Time	rule.id	rule.description	agent.ip	predecoder.program_name	input.type
> Dec 9, 2025 @ 11:01:19.590	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 11:00:33.475	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 11:00:11.352	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 10:57:39.387	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
Dec 9, 2025 @ 10:56:45.000	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log

```
(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret services
is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-12-09 20:22:14
[DATA] max 4 tasks per 1 server, overall 4 tasks, 132 login tries (l:11/p:12), ~33 tries per task
[DATA] attacking ssh://192.168.50.10:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://medsecureadmin@192.168.50.10:22
[ERROR] could not connect to ssh://192.168.50.10:22 - Timeout connecting to 192.168.50.10

(kali@kali)-[~]
$ ping 192.168.50.10
PING 192.168.50.10 (192.168.50.10) 56(84) bytes of data.
^C
— 192.168.50.10 ping statistics —
5 packets transmitted, 0 received, 100% packet loss, time 5185ms
```


PHASE 3



< Manager configuration

Edit ossec.conf of Manager

```
<command>  
  <name>netsh</name>  
  <executable>netsh.exe</executable>  
  <timeout_allowed>yes</timeout_allowed>  
</command>
```

```
<active-response>
  <disabled>no</disabled>
  <command>firewall-drop</command>
  <location>local</location>
  <rules_group>sshd</rules_group>
  <timeout>0</timeout>
</active-response>
```

FINAL FINDINGS & **IMPACT**

Weak passwords and excess user privileges found.

Detected SSH brute-force, scanning, and suspicious access.



Risks included data exposure and compliance issues.



Correlated logs confirmed real Indicators of Compromise.



To make the system safer, start by fixing technical issues like securing SSH access, closing unnecessary ports, and improving firewall rules

1

Improve firewall rules and extend log retention.

2

regularly update and patch systems, redesign the network to protect important assets, and train staff on security best practices.

3

RECOMMENDATION



Wazuh proved effective for detection
and response.

1

Project mirrored real SOC procedures.

2

Key takeaway: continuous monitoring
and strong processes are critical.

3

CONCLUSION



A top-down view of a desk with various items: a laptop in the upper center, a cup of coffee in the upper left, a pen in the center, a pair of glasses in the lower center, several paper clips on the left, and a large green leaf in the bottom left corner. The background is a light, neutral color.

THANK YOU